

Coding Challenge:

1. A cryptanalyst is tasked with decoding a sequence of two lines of digits using specific rules. The digits represent letters, and the relationship between the digits and the letters must be determined based on the given constraints. Your task is to write a Python program that translates the digits into letters according to the rules below.

Given:

- Two lines of digits:
 - Line 1: 933456667
 - Line 2: 223344578

Rules:

1. Each of the digits from 2 to 9 represents exactly one of the eight letters: A, E, I, O, U, R, S, T.
2. Each letter is represented by exactly one of the digits.
3. If a digit occurs more than once, it always represents the same letter.
4. The following letter frequency constraints are known:
 - The letter T and the letter O are each represented exactly 3 times.
 - The letter I and the letter A are each represented exactly 2 times.
 - The letter E is represented exactly 4 times.
 - The letters U, R, and S appear once each and are represented by the remaining digits.

Objective:

Write a Python program that decodes the two lines of digits into corresponding letters based on the above constraints.

Output:

- The program should output two decoded lines where each digit is replaced by its corresponding letter.

Example Input:

Line 1: 933456667
Line 2: 223344578

Example Output:

Decoded Line 1: OETARSSTT
Decoded Line 2: IIAEOTSRU

2. Write a Java program to read Student data (Name, Age, Gender, Marks), Sort the data as per their marks and age (if marks are same, age should be consider for sorting)

- i) Sorting based on Age to be ascending (low-high) order**
 - ii) Sorting based on Marks to be descending (high-low) order**
- Sample File to be read as below

Name, Age (YYYY-MM-DD), Gender (Male/Female), Marks

Prakash, 1984-12-13, Male, 90.00 Manoj, 1981-07-14, Male, 90.00

Shekhar, 1982-04-27, Male, 80.0 Archana, 1979-09-28, Female, 91.9 Sonali, 1992-02-01, Female, 91.9

Devashish, 1984-05-23, Male, 92.0

Amit, 1984-09-10, Male, 92.0

SQL Challenge:

Consider following schema and write sql queries for following scenario:

department

Dep_id INTEGER

dep name VARCHAR(20)

Dep_location INTEGER

VARCHAR(15)

employees

Emp_id INTEGER

emp name VARCHAR2(15)

Job_name VARCHAR2(15)

Manager_id INTEGER

hire_date DATE

Salary DECIMAL(10,2)

Commission DECIMAL(7,2)

Dep_id INTEGER

1. Select unique job names from employees whose department is named as Engineering and whose name starts with "Dav".
2. List all employee names along with their manager names.
3. List all employees who have the highest salary in their own department.
4. List all non-manager employees who have the highest salary in their own department.
5. Get the count of the number of employees having more than 3 years of experience.
6. Get the count of all employees working at the Pune location.
7. List 3 employees whose salary is more than \$20,000
8. Get the name of the department which has the highest expenditure on salaries.

Linux Challenge:

- Sort contents of file /tmp/cities.txt in reverse order
- List all environment variables whose name contain "ORACLE" substring
- List all files/directories in /tmp on Last Modified Time (In Reverse Order)
- List all running java processes on the system
- List all processes in the system
- Kill a process using its process i

MCQ:

1. Which cases would you use an outer join?

- a) The tables being joined have NOT NULL columns.
- b) The tables being joined have only matched data.
- c) The columns being joined have NULL values.
- d) The tables being joined have only unmatched data

2. An alternate key

- a) Can have duplicate values
- b) Is another name of foreign key
- c) Is the second column of the composite primary key
- d) None of the above

3. Which of the following SQL features is NOT supported by MySQL?

- a) UNION ALL
- b) CHECK Constraint
- c) Materialized Views
- d) All of the above

4. A subquery can NOT be used within another -

- a) INSERT query
- b) UPDATE query
- c) DELETE query
- d) None of the above

5. Consider the below given table schemes

emp (empno, ename, job, sal, deptno} dept (deptno, dname)

Which of the following query will be used to retrieve all employees (ename) with their respective departments (dname) irrespective of whether a department is assigned to the employee or not?

- a) **SELECT ename, dname FROM emp e JOIN dept d ON (e.deptno = d.deptno);**
- b) **SELECT ename, dname FROM emp e LEFT OUTER JOIN dept d ON**

(e.deptno = d.deptno); c) SELECT ename, dname FROM emp e RIGHT OUTER JOIN dept d ON (e.deptno = d.deptno);

d) SELECT ename, dname FROM emp e FULL OUTER JOIN dept d ON (e.deptno = d.deptno);

6. If EMP relation contains 15 rows and DEPT relation contains 10 rows. How many rows will be returned by Cartesian Join of EMP and DEPT relations?

- a) 25
- b) 150
- c) 50
- d) No row will be returned

7. What kind of relationship exists between the following sets of entities?

- i) Between Person and AadhaarCard
- ii) Between Person and House
- a) (i) One to One, (ii) Many to Many
- b) (i) One to One, (ii) Many to One
- c) (i) One to Many, (ii) One to Many
- d) (i) Many to Many, (ii) One to Many

9. Which of the followings are true in context of relational model:

- i) All metadata information about the tables are stored separately in metadata files
- ii) Each tuple or row must be uniquely identifiable
- iii) The tuples are stored in ordered on the basis of the primary key of the relation/table

- A. Only i is true
- B. Both i and ii are true
- C. Only ii is true
- D. Both ii and iii are true

10. Which of the following query will return records of all employees from EMP-table having 'h' as 2nd character and 'a' as the last character in their name?

- A. SELECT FROM EMP WHERE employee_name LIKE '%h%a'
 - B. SELECT FROM EMP WHERE employee_name LIKE '_ha'
 - C. SELECT FROM EMP WHERE employee_name LIKE 'h'
 - D. SELECT FROM EMP WHERE employee_name LIKE '_h_a'
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Aptitude:

1. Walk a distance of 16 km from his house to his friend's house, He walks at a speed of 8 km/hr and after every 30 minutes he takes rest for 10 minutes. How much time does a man have to walk to reach his friend's house?
 - a) 2 hours 10 minutes
 - b) 2 hours 30 minutes
 - c) 4 hours 40 minutes
 - d) 4.2 hours 20 minutes
2. Ajay works in a certain v 16 days. He starts the work, works for 4 days and then quits. Then Bharat takes up the job and does the remaining work. If Bharat alone takes 24 days to do the entire work, in how many days will the work be completed?
 - a) 122 days
 - b) 24 days
 - c) 26 days
 - d) 28 days
3. P and Q can do a certain amount of work in 28 days and 56 days respectively. P works for 7 days and then Q joins P. In how many more days can they complete the work?
 - 1. 7 days
 - 14 days

3. 21 days

4. 28 days

4. ZA5, VB, XC6, W3D,

A E7V

B.V2E

C.VE7

D VES

5. Alfred buys an old scooter for Rs. 4700 and spends Rs. 800 on its repairs. If he sells the scooter for Rs. 5800, his gain percent is:

A. 4.57%

B.45%

C 10%

D. 12%

6. A and B can do a work in 60 days; B and C can do it in 120 days; A and C can do it in 80 days. At what time A alone can do the work?

A. 104

B. 96

C. 80

D. 70

7. PETAL: FLOWER

A salt: pepper

B. tire: bicycle

C base: ball

D sandals: shoes

E. puppy: dog

8. Statements: All buildings are chalks. No chalk is toffee.

Conclusions:

No building is toffee

All chalks are buildings.

A Only (1) conclusion follows

B. Only (2) conclusion follows

C Either (1) or (2) follows

D. Neither (1) nor (2) follows

E. Both (1) and (2) follow

9. The present ages of A, B and C is 92 years. If 4 years ago, the ratio of their ages were 1:2:3 respectively, find A's present age.

A. 18.5 year

B. 14.8 years

17.3 years

D. 20.3 years

10. If 20% of $a = b$, then $b\%$ of 20 is the same as:

A 4% of a

B. 5% of a

C. 20% of a

D. None of these